

POINT ESTIMATION POWERFUL ULTIMATE SENTENCE (PEPUS)

CASES	θ	$\hat{\theta}$	se: standard error estimate
$X_1, X_2, \dots, X_n$ i.i.d. $E[X_i] = \mu$ $Var(X_i) = \sigma^2$	1 μ	$\bar{X} \sim N(\mu, \frac{\sigma^2}{n})$ $SE: \sigma/\sqrt{n}$	$\frac{s}{\sqrt{n}}$
$X_1, X_2, \dots, X_n$ i.i.d. $X_i \sim \text{Bernoulli}(p)$ $(E[X_i] = p, Var(X_i) = p(1-p))$	2 p	$\hat{P} \sim N(p, \frac{p(1-p)}{n})$ $SE: \sqrt{\frac{p(1-p)}{n}}$	$\sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$
$X_1, \dots, X_{n_x}$ i.i.d. $E[X_i] = \mu_x$ $Var(X_i) = \sigma_x^2$ $Y_1, \dots, Y_{n_y}$ i.i.d. $E[Y_j] = \mu_y$ $Var(Y_j) = \sigma_y^2$ x, y independent	3 $\mu_x - \mu_y$ (μ_d)	$\bar{X} - \bar{Y} \sim N(\underbrace{\mu_x - \mu_y}_{=: \mu_d}, \frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y})$ $SE: \sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}$	$\bar{x} - \bar{y}$ If $\sigma_x^2 \neq \sigma_y^2$: $\sqrt{\frac{s_x^2}{n_x} + \frac{s_y^2}{n_y}}$ If $\sigma_x^2 = \sigma_y^2$: $\sqrt{\frac{s_{pool}^2}{n_x} + \frac{s_{pool}^2}{n_y}}$

$$s_{pool}^2 = \frac{(n_x - 1)s_x^2 + (n_y - 1)s_y^2}{n_x + n_y - 2}$$

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CASES	θ	$\hat{\theta}$	$\hat{\theta}$	se: standard error estimate
$(x_1, y_1), \dots, (x_n, y_n)$ $D_1, D_2, \dots, D_n$ i.i.d. $E[D] = \mu_x - \mu_y$ $\text{Var}(D) = \sigma_x^2 + \sigma_y^2 - 2\sigma_{xy}$ $= \sigma_D^2$ X, Y PAIRED $D_i = X_i - Y_i$	4 $\mu_x - \mu_y$ (μ_D)	$\bar{D} = \bar{X} - \bar{Y} \sim N\left(\mu_x - \mu_y, \frac{\sigma_D^2}{n}\right)$	$\bar{x} - \bar{y}$ ($\bar{D}$)	$\frac{s_D}{\sqrt{n}}$
$X_1, \dots, X_{n_x}$ i.i.d. $X_i \sim \text{Bernoulli}(p_x)$ $Y_1, \dots, Y_{n_y}$ i.i.d. $Y_i \sim \text{Bernoulli}(p_y)$ X, Y independent	5 $p_x - p_y$	$\hat{P}_x - \hat{P}_y$ $\sim N\left(p_x - p_y, \frac{p_x(1-p_x)}{n_x} + \frac{p_y(1-p_y)}{n_y}\right)$	$\hat{P}_x - \hat{P}_y$	$\sqrt{\frac{\hat{P}_x(1-\hat{P}_x)}{n_x} + \frac{\hat{P}_y(1-\hat{P}_y)}{n_y}}$

CASE 1: EXAMPLE:

SURF II Estimate the population mean duration μ of calls at a call centre

- $X_1, X_2, \dots, X_{300}$ i.i.d. calls $E[X_i] = \mu, \text{Var}(X_i) = \sigma^2$
- We use the sample mean estimator $\bar{X}$ (recall $E[\bar{X}] = \mu, \text{Var}(\bar{X}) = \frac{\sigma^2}{n}$)
- The sample mean estimate corresponding to the sample is $\bar{x} = 254.3733$ which is our estimate of μ
- If σ is known ($\sigma = 150$), the standard error of $\bar{X}$ is $SE = \frac{\sigma}{\sqrt{n}} = \frac{150}{\sqrt{300}} = 8.6602$
- If σ is not given, the estimate for the standard error of the estimator is

$$se = \frac{\overset{\text{sample standard deviation}}{\textcircled{5}}}{\sqrt{n}} = \frac{162.85}{\sqrt{300}} = \boxed{9.7376}$$

The estimate of the standard error se is the estimate of the expected distance of the estimator $\bar{X}$ from the parameter μ .

It does not tell us anything about the distance of our specific estimate $\bar{x} = 254.37$ from the parameter μ .

CASE 2 EXAMPLE: Estimate the population proportion p of solved cases sum 17 at a call centre

- $X_1, \dots, X_{300}$ i.i.d. cases $X_i \sim \text{Bernoulli}(p)$ ($X_i = 1$ if case i solved)
 - Use the sample proportion estimator $\hat{P} \sim N(p, \frac{p(1-p)}{n})$
 - The sample proportion estimate corresponding to the sample is $\hat{p} = \frac{273}{300} = 0.91$ which is an estimate of the parameter p .
 - The standard error of the estimator is $\sqrt{\frac{p(1-p)}{n}}$, but p is unknown
 - The estimate of the standard error is $se = \sqrt{\frac{\hat{p}(1-\hat{p})}{n}} = 0.01652$
- estimate of the expected distance of the estimator from the parameter of interest

CASE 3 EXAMPLE: Estimate the difference in average refund to smokers and non smokers $\mu_x - \mu_y$

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• $X_1, \dots, X_{115}$ i.i.d. refunds to smokers. $E[X_i] = \mu_x$, $\text{Var}(X_i) = \sigma_x^2$

$Y_1, \dots, Y_{459}$ i.i.d. refunds to non smokers. $E[Y_i] = \mu_y$, $\text{Var}(Y_i) = \sigma_y^2$

• Use the estimator $\bar{X} - \bar{Y}$

• Corresponding to our sample $\bar{x} = 31341$ $\bar{y} = 7612$

our estimate for $\mu_d = \mu_x - \mu_y$ is $\bar{x} - \bar{y} = 23730$ (in the sample on average smokers have refund 23730 higher!)

independent samples

• $\text{Var}(\bar{X} - \bar{Y}) \stackrel{\uparrow}{=} \text{Var}(\bar{X}) + \text{Var}(\bar{Y}) = \frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y} \longrightarrow \text{STANDARD ERROR} = \sqrt{\frac{\sigma_x^2}{n_x} + \frac{\sigma_y^2}{n_y}}$

$\sigma_x^2 \neq \sigma_y^2$

$$S_{\text{POOL}}^2 = \frac{(n_x - 1)S_x^2 + (n_y - 1)S_y^2}{n_x + n_y - 2}$$

ESTIMATE STANDARD ERROR

$$\sqrt{\frac{S_x^2}{n_x} + \frac{S_y^2}{n_y}} = 1115$$

$\sigma_x^2 = \sigma_y^2$

ESTIMATE STANDARD ERROR

$$\sqrt{\frac{S_{\text{POOL}}^2}{n_x} + \frac{S_{\text{POOL}}^2}{n_y}} = 768$$

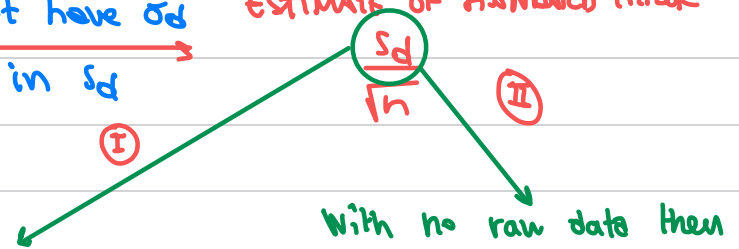
CASE 4 EXAMPLE: Estimate the difference in mean cholesterol level before and after a drug $\mu_d = \mu_x - \mu_y$

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- $(X_1, Y_1), (X_2, Y_2), \dots, (X_n, Y_n)$ pairs of measurements. Hence: $D_1, D_2, \dots, D_n$ i.i.d. differences in cholesterol levels. $D_i = X_i - Y_i$
 - Use as estimator $\bar{X} - \bar{Y} = \bar{D}$
 - Corresponding to our sample $\bar{x} = 155$ $\bar{y} = 122$
- Our estimate of $\mu_d = \mu_x - \mu_y$ is $\bar{x} - \bar{y} = 33 = \bar{d}$
- Here the samples are PAIRED! Defo not INDEPENDENT!

• $\text{Var}(\bar{X} - \bar{Y}) = \text{Var}(\bar{D}) = \frac{\sigma_d^2}{n}$ we don't have σ_d ESTIMATE OF STANDARD ERROR

$\xrightarrow{\text{we plug in } s_d}$



If you have the raw data,
 s_d is the standard deviation
 of the differences: $x_1 - y_1, x_2 - y_2, \dots, x_n - y_n$
 $s_d = s_d(d)$

$s_d = \sqrt{s_x^2 + s_y^2 - 2s_{xy}}$

CASE 5 EXAMPLE: Estimate the difference in proportion of clicks between banner A and banner B $p_A - p_B$

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- $X_1, \dots, X_{2364}$ i.i.d. views of banner A. $X_i \sim \text{Bernoulli}(p_A)$ $X_i = 1$ if person i clicked on the banner
- $Y_1, \dots, Y_{2323}$ i.i.d. views of banner B. $Y_i \sim \text{Bernoulli}(p_B)$ $Y_i = 1$ if person i clicked on the banner

- Use estimator $\hat{p}_A - \hat{p}_B$

- Corresponding to our sample $\hat{p}_A = 0.1929$ $\hat{p}_B = 0.2630$

Our estimate of $p_A - p_B$ is $\hat{p}_A - \hat{p}_B = -0.0701$

- $\text{Var}(\hat{p}_A - \hat{p}_B) \stackrel{\text{independent samples}}{=} \text{Var}(\hat{p}_A) + \text{Var}(\hat{p}_B) = \frac{p_A(1-p_A)}{n_A} + \frac{p_B(1-p_B)}{n_B}$

ESTIMATE OF STANDARD ERROR: $se = \sqrt{\frac{\hat{p}_A(1-\hat{p}_A)}{n_A} + \frac{\hat{p}_B(1-\hat{p}_B)}{n_B}} = 0.012219$

